

Scientific Research is a Token of Humankind's Survival

- Vladimir Keilis-Borok

- Understanding the text

* Answer the following Questions:

a) What does a scientist get instead of big money?

Ans: A scientist gets freedom, camaraderie and independence instead of big money, as explained by 'Vladimir Keilis Borok' in his essay. Science is an exciting adventure where the major reward comes from the discovery itself as scientists are rather more interested in discoveries that serve the mankind than making big money. Honours and promotions depend on hard work more than in other occupations. Despite lower earning than businessman, lawyers ~~and~~ or doctors, scientists enjoy intellectual satisfaction, global friendships and autonomy is ~~per~~ pursuing truth to safeguard humanity's future.

b) What was the problem that the nuclear powers had faced?

Ans: Nuclear power faced the critical challenges of reliably detecting small underground

nuclear explosions and distinguishing them from natural earthquakes to verify and test ban treaty without on-site inspections which could compromise security. This verification barrier threatened arms control efforts and heightened global risks of undetected violations escalating conflicts. Scientists tackled it through advanced seismology developing precise methods for identification fostering international trust and cooperating proving research essential for survival by enabling enforceable agreements that curb destructive capabilities and protect humanity from nuclear catastrophe.

c) In which area did Keilis Borok's theoretical knowledge have a direct application?

Ans: Keilis Borok's theoretical knowledge in seismology had direct application in the critical area of distinguishing small underground nuclear explosions from natural earthquakes enabling reliable verification of test ban treaties without intrusive inspections. This breakthrough addressed nuclear powers' verification problems fostering international trust and arms control cooperation. His research ~~emphasized~~ exemplified how pure science contributes to human kind's survival by resolving barriers to

peace providing objective methods that prevent undetected violations and reduce global ~~con~~ catastrophe risks through collaborative intellectual efforts.

d) What was the important decision that the politicians took before Geneva Summit?

Ans Before the Geneva Summit, politicians from the nuclear powers made the crucial decision to pause their political rivalry and turn to technical experts. They realized that a nuclear test ban treaty was only possible if it could be verified. Therefore they decided to convene a group of scientists to determine if it was scientifically possible to detect secret underground nuclear tests and distinguish them from earthquakes.

e) Natural and man made disasters.

Ans The author mentions several types of disasters that threaten human survival. Natural disasters include massive earthquakes, tsunamis, and self-inflicted environmental change. Man-made disasters include the catastrophic threat of Nuclear war and the potential for a "ticking time bomb" of social or environmental collapse. The author argues that science is our best tool to predict, or manage these immense risks to our civilization.

• Reference to the content

a) The professional addressed as 'you' in the sentence "If you are clever, why are you so poor?" refers to a ---.

Ans The professional addressed as 'you' is
i) Scientist

In this essay, the author explains that people often judge success by how much money someone makes. Since Scientists usually earn much less than a Businessman or lawyers, they are often asked why they aren't rich if they are supposedly so "clever."

The author justifies this by saying that scientists have a different set of values. Instead of chasing a high salary, they are motivated by the excitement of uncovering the secrets of nature and the honor of being recognized by their peers worldwide.

b) The writer says, "I found myself in Geneva." What does it express?

Ans This phrase describes a moment of sudden realization and surprise. For years, the author lived in the quiet world of theoretical science, focusing on complex math and earthquake pattern. He never expected his "ivory tower" research to have any

impact on the real world or international politics.

Suddenly, because of the cold war, his expertise was needed to solve a massive political problem: how to detect secret underground nuclear tests. This move to Geneva represents the moment he stepped out of his laboratory and into the spotlight of global diplomacy, where his work could actually help prevent a war.

It shows that science is not just a lonely hobby; it is a bridge that connects different nations. Being "in Geneva" symbolizes how his intellectual skills gave him a seat at the table with world leaders, providing that scientific knowledge is a universal language that can help save the world.

c) Are the following statement true? why or why not? Discuss with your partner.

is Money is more powerful intellectual resources.
 my No the statement is false. According to the essayist, money is not more powerful than intellectual resources. Scientists earn less than businessman or doctors yet their profession provides practical solutions for survival like nuclear verifications and disaster

prediction provides intellectual efforts surpass monetary power in safeguarding humanity.

ii) Intellectual resources help survival of the mankind.

Ans Yes, the statement is true. Intellectual resources through scientific research are vital for mankind's survival. They enable detection of nuclear tests, prediction of earthquakes and tsunamis, and resolution of global threats, offering objective knowledge and international trust that protect civilization from catastrophes.

iii) Basic research is a way of stalling disasters.

Ans Yes, the statement is true. Basic research is a way of stalling disasters as it develops tools to forecast natural calamities like earthquakes and prevents man-made ones by verifying arms control ~~the~~ treaties, thereby reducing risks of nuclear war and environmental crises for humanity's long term existence.

d) How does the essayist justify that scientific research is the humankind's survival?

Ans. The essayist justifies that scientific research is humankind's survival by showing its power to solve critical global threats.

Through international scientific ~~team~~ teamwork, seismologists devised accurate ways to tell apart small underground nuclear explosions from earthquakes overcoming verification obstacles for test ban treaties and fostering trust among rival nations.

This achievement helped curb the arms race and avert nuclear catastrophe while applying to predict natural disasters such as earthquakes, tsunamis and typhoons. Science delivers objective, reliable solutions where political efforts often fall short.

In essence, it grants freedom, global camaraderie, and vital discoveries that safeguard civilization from both man-made and natural perils, making research essential for humanity's continued existence.

Q. What can be the purpose of the essayist using quotations in the essay?

Ans. The essayist uses quotations to add authenticity and authority to his arguments on science as humankind's survival token. By citing experts, politicians, and key statements from Geneva talks, he strengthens the narrative of scientific collaborations solving nuclear verification issues with precision.

Quotations vividly illustrate real-world challenges like distinguishing explosions from earthquakes and highlight the shift from rivalry to trust. They make abstract ideas concrete and emotionally resonant for readers. It provides vivid descriptive understanding of complex scientific problems and methods which enables even the scientifically poor people to easily grasp the essayist's intentions.

This technique bridges personal experience with broader historical context, emphasizing science's practical role in averting disasters and fostering global peace through objective research.

f) Discuss and illustrate the writer's stand that scientists are the most practical people in the world.

Ans The essayist asserts that scientists are the most practical people in the world because their theoretical work delivers tangible solution to humanity's gravest threats. Unlike businessmen or politicians driven by profit or power, scientists apply pure research directly to real problems, such as developing seismological methods to distinguish underground nuclear explosions from earthquakes for test ban verification.

This practicality enables nuclear powers to build trust and curb the arms race without risky inspections. Their efforts extend to predicting disaster like the earthquakes, tsunamis and others, proving science's immediate utility in safeguarding lives and civilization.

By prioritizing objective truth, independence and global collaboration over monetary rewards, scientists emerge as true pragmatists ensuring long-term human survival amid escalating perils.